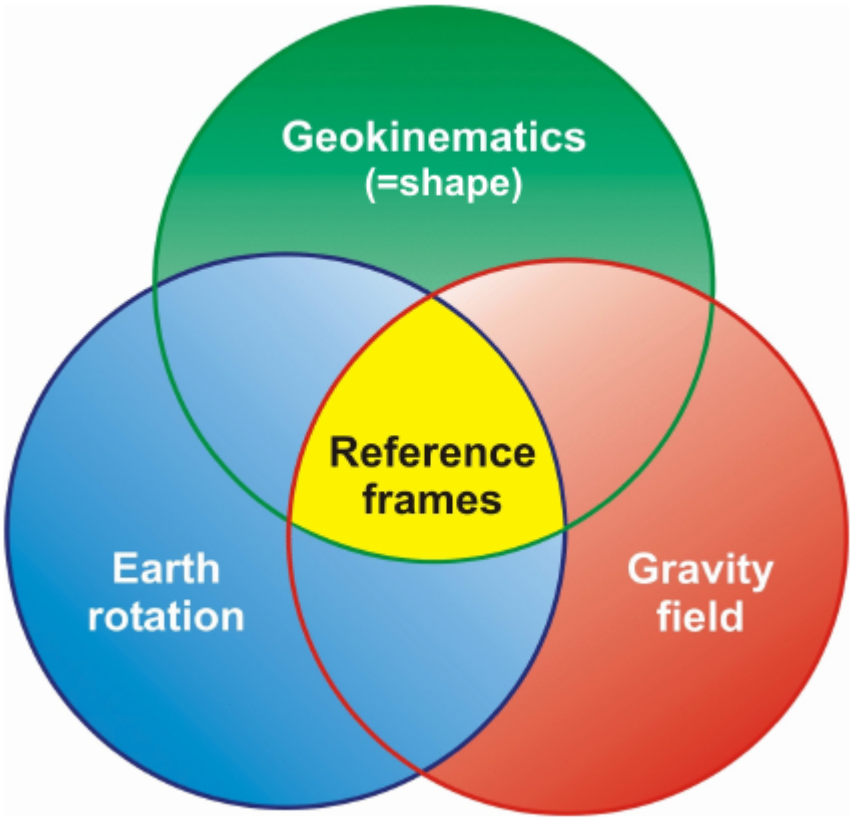
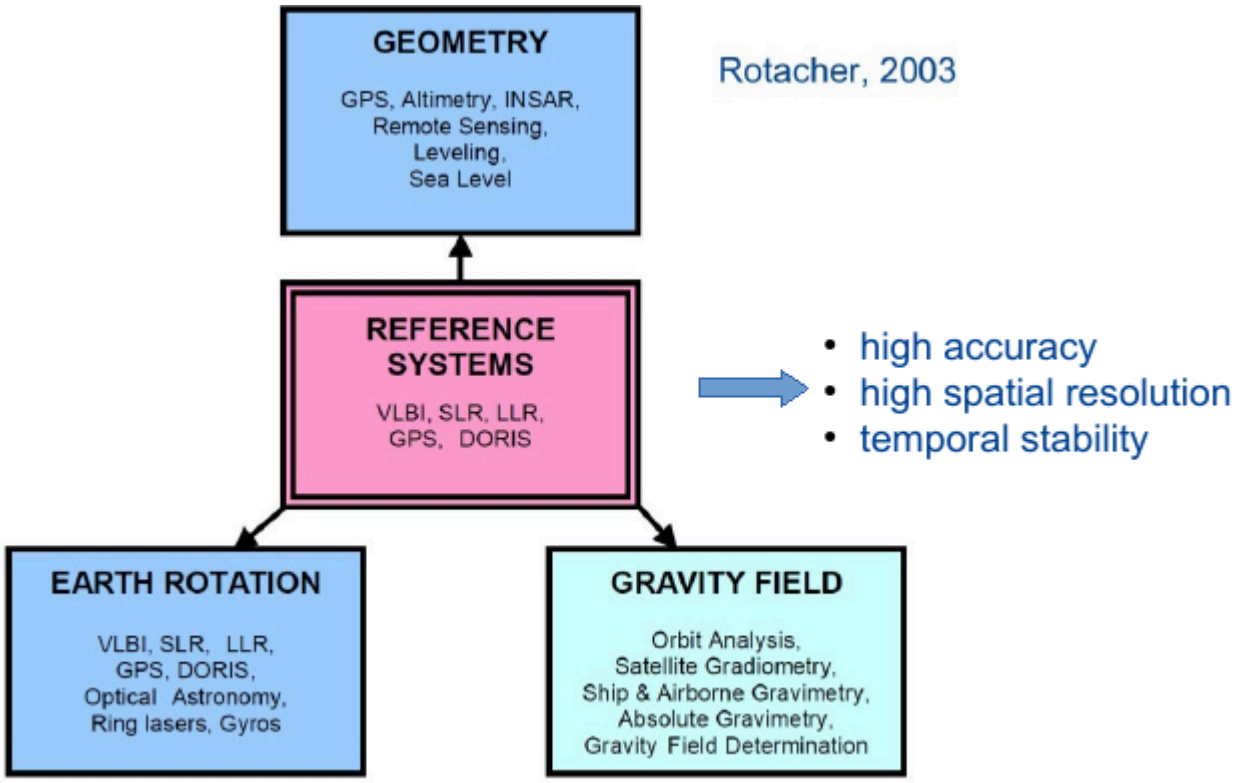
Lecture 1: Introduction - Basics of Space Geodetic Observing Systems & Reference Ellipsoid Concept

Geodesy:
Science concerned with (Friedrich Robert **Helmert** (1880))

- Study of the **shape and size** of the Earth in the **geometric sense**
- Form of the **Equipotential surfaces** of the gravity potential
- (still relevant and poineer)

3 Pillars of Geodesy





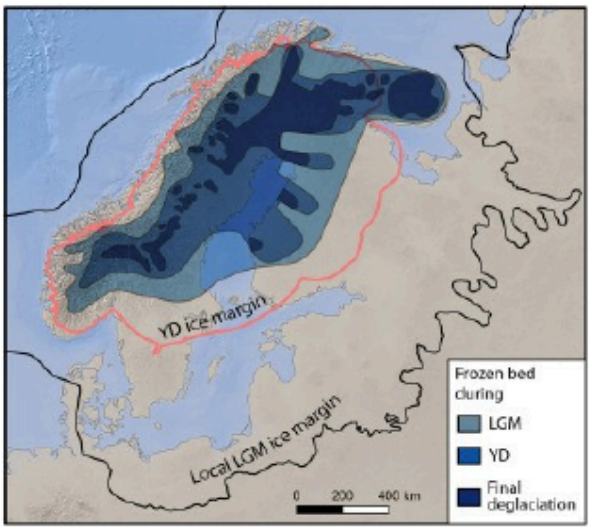
Practical consideration of Markers:

- Stability
- Visibility
- Accessibility

Equipotential Surfaces

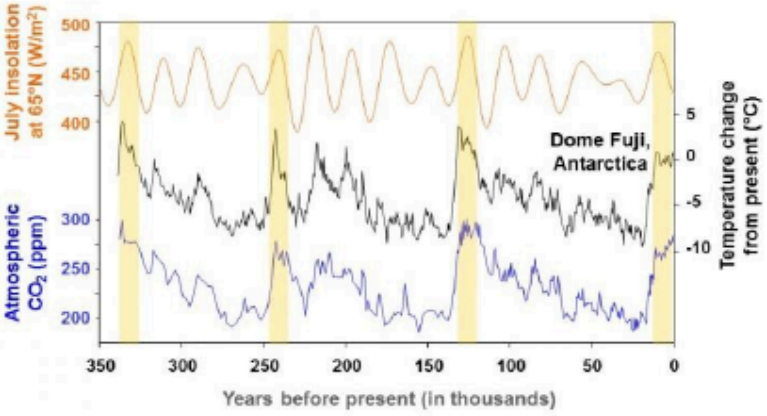
- A special equipotential surface: geoid
  - Surface of a hypothetical ocean at rest
  - Makes sense as a natural reference surface

Ice Ages



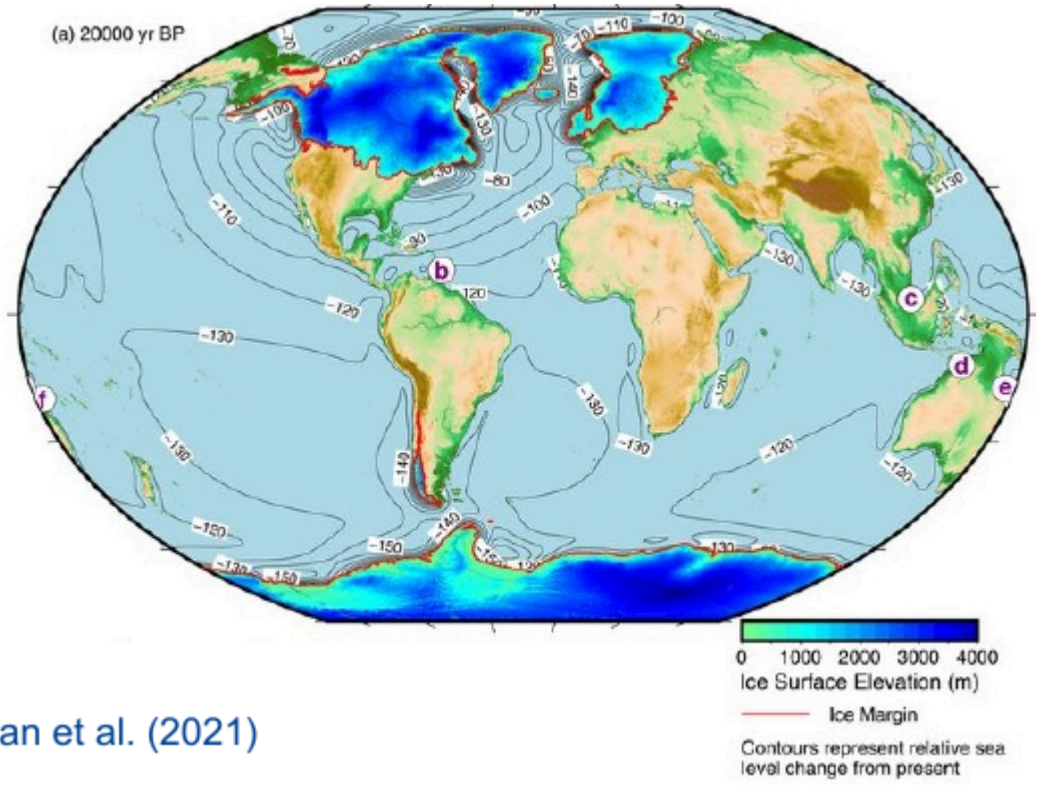
Stroeve et al 2016

- Ice shields come and go in ~100000 year cycles



noaa.gov

Glacial and Deglaciation periods

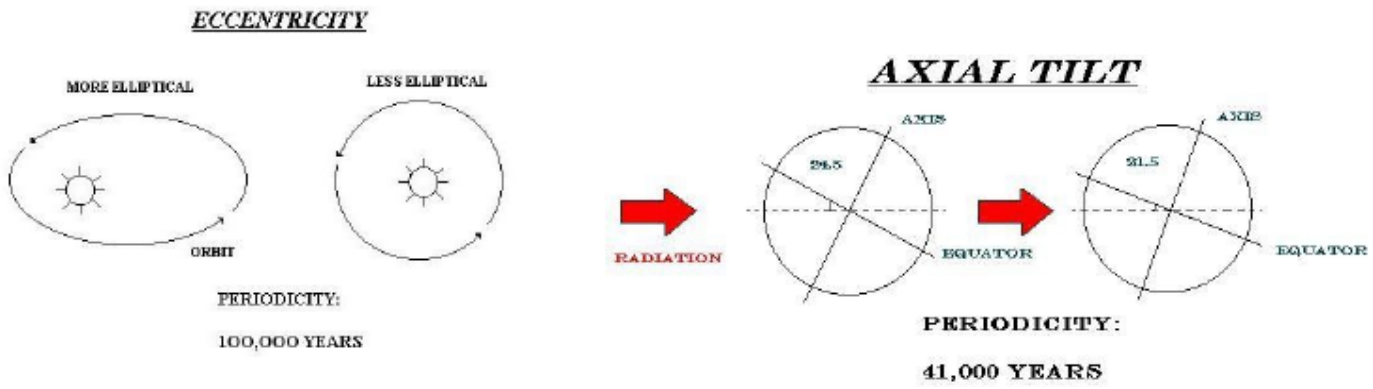


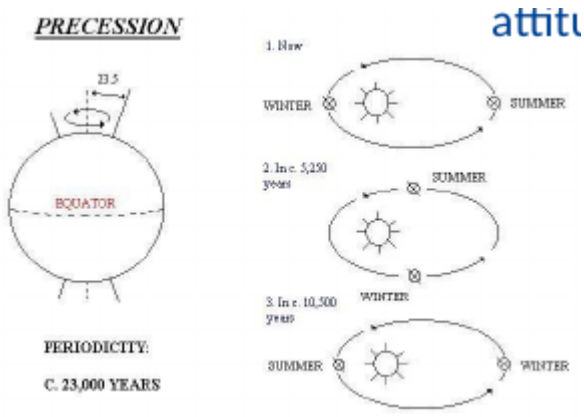
Gowan et al. (2021)

Ice Ages and Earth Rotation

- Milankovitch Cycles: Slowly changing attitude and orbit of Earth around the sun
  - Eccentricity (100,000 years)
    - Changes in Earth's orbital shape (circular vs. elliptical)

        - More elliptical -> seasonal effects from precession become stronger -> glacial-interglacial cycles
    - Axial tilt (Obliquity) (41,000 years)
        - Angle of Earth's axis (22.1°-24.5°)
        - Change of Seasonal Contrast
    - Aphelion precession (23,000 years)
        - Timing of Aphelion (Earth farthest from Sun) and Periphelion shifts relative to season
        - Change of season in which Earth is closest or farthest from Sun
    - Axial precession (26,000 years)
        - Wobble of Earth's axis
        - Direction of rotation axis changes



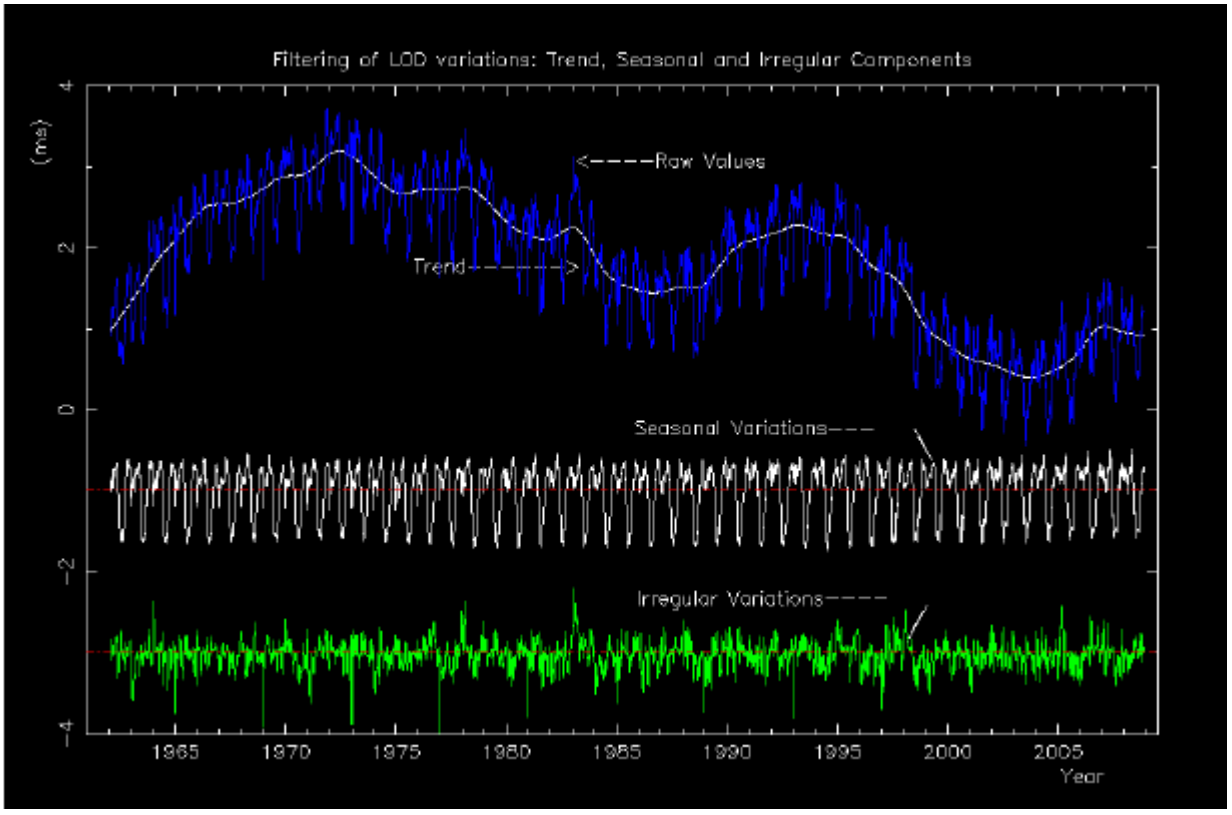


- Axial and Aphelion Precession -> 23k year climate cycle

Cycle	Period	What Changes	Climate Impact
Aphelion precession	~23,000 yrs	Timing of closest/farthest Sun distance vs seasons	Alters seasonal strength
Axial precession	~26,000 yrs	Direction Earth’s axis points	Shifts seasonal timing
Eccentricity	~100,000 yrs	Shape of Earth’s orbit	Controls strength of other cycles

Cycle	Period	What Changes	Climate Impact
Axial tilt	~41,000 yrs	Angle of Earth's axis	Controls seasonal intensity

Length of Day

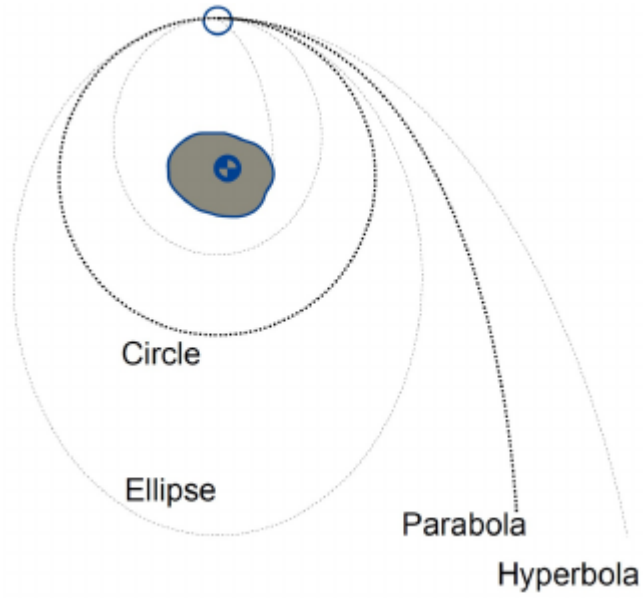
- Length of day changes over time
- Different geophysical effects are superimposed
- 

- In space: Special and General relativity

Global Reference System

- Essentially *cake recipe*
  - Origin: Long-term Mean center of mass
    - Center of figure
  - Describes Earth's rotation axis over time relative to the stars
  - For tectonic plate motion
    - refer to volcanic hotspots (underlying mantle)
    - restrict: no-net-rotation of all plates
  - ...
- Resulting Frame is the actual cake
  - Requires geodetic observations

- Subject to failures and uncertainties



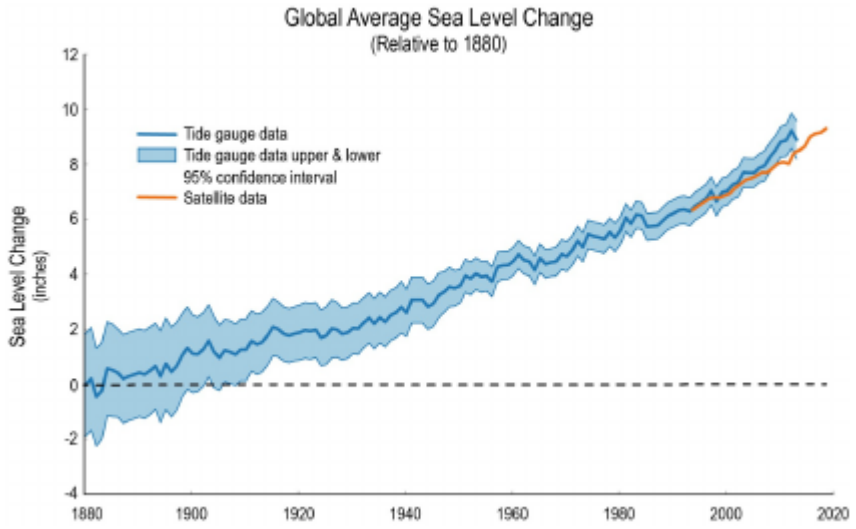
Orbits:

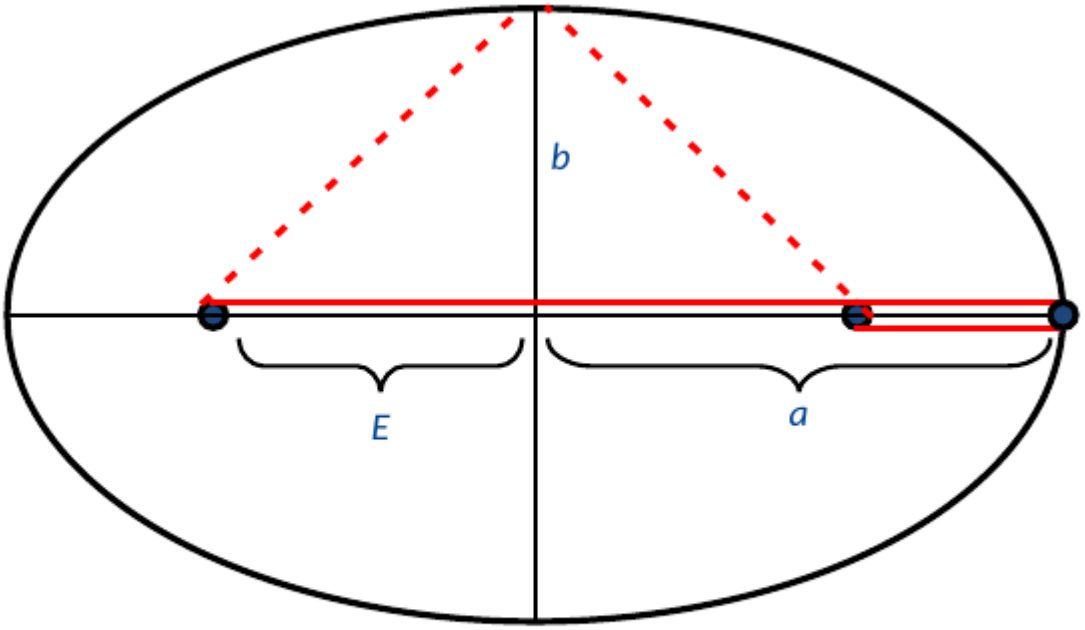
- Repeat orbits
- sun-synchronous orbits

Monitoring the Ocean

Applications of satellite altimetry:

- Ocean climate monitoring
- Large lakes and rivers
- Ocean tides, currents, eddies
- Bathymetry, sea ice
- Windstress, wave height



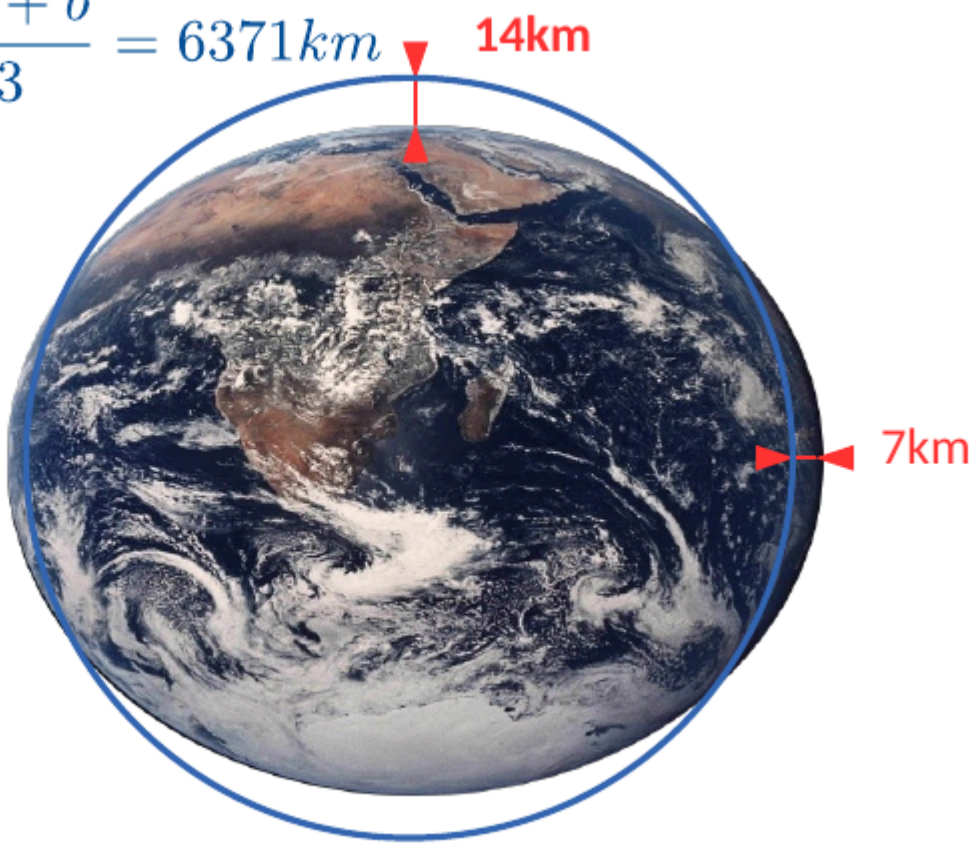


Given the geometry of the red triangle, what would be the linear Eccentricity E?

$$E = \sqrt{a^2 - b^2}$$

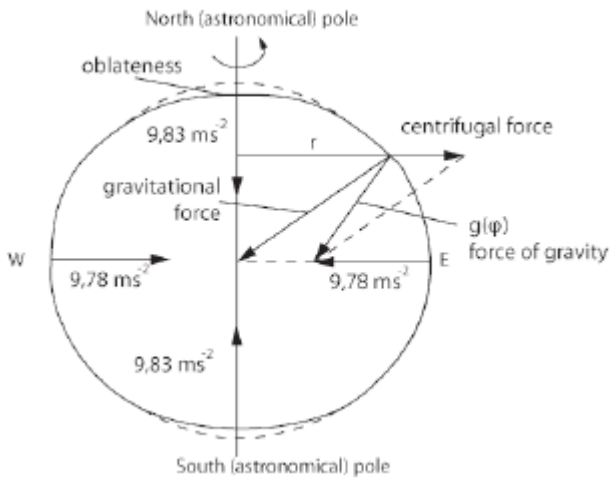
Name	Parameter	Earth
Semi-Major Axis	a	6378.137 km
Semi-Minor Axis	b	6356.752 km
Linear eccentricity	$E = \sqrt{a^2 - b^2} = ea$	521.854 km
First eccentricity	$e = \frac{E}{a}$	0.0818 …
Second eccentricity	$e' = \frac{E}{b}$	0.0821 …
Oblateness	$f = (a - b)/a$	0.003 352 ..

$$r^*_{mean} = \frac{2a+b}{3} = 6371km$$



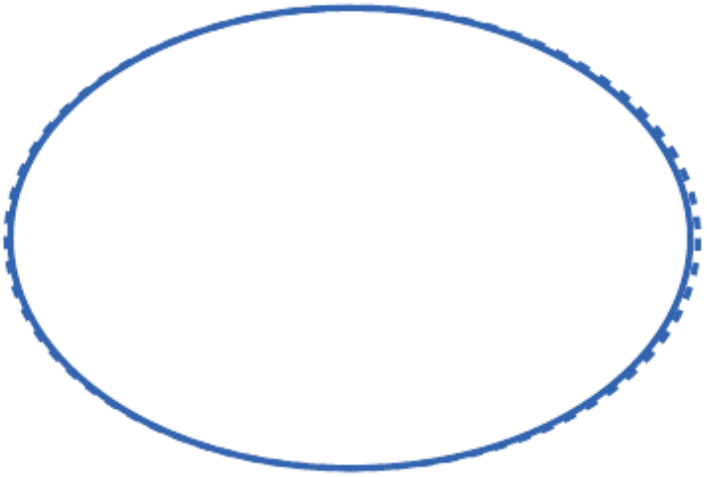
* Arithmetic mean, but other types of means possible, e.g. radius of sphere with equivalent surface area or volume

Earth Rotation

- Due to Earth's rotation we experience a centrifugal force
  - Depends on latitude
  - Earth rotation
- Mean Earth rotation $\rightarrow \frac{2\pi}{\text{sideral day}}$
  - Rotation relative to the stars
- Ellipsoidal Earth Model
  - Add mean Earth rotation as additional defining parameter
- 

Gravity field of Ellipsoidal Earth Model

- Define total gravitational mass of the Ellipsoid (M)
- In addition to the shape: the gravity field is also chosen to be ellipsoidal! → another gravity parameter needed (dynamical form factor J_2)
- Amount of parameters for the ellipsoidal model so far: 5 (a, b, M, omega, J2)
- Additional constraint:
  The geometrical surface coincides with the reference 'equipotential' surface (depends on density and volume)
  → introduces dependency
  → 4 Defining parameters are needed (a, M, omega, J_2)



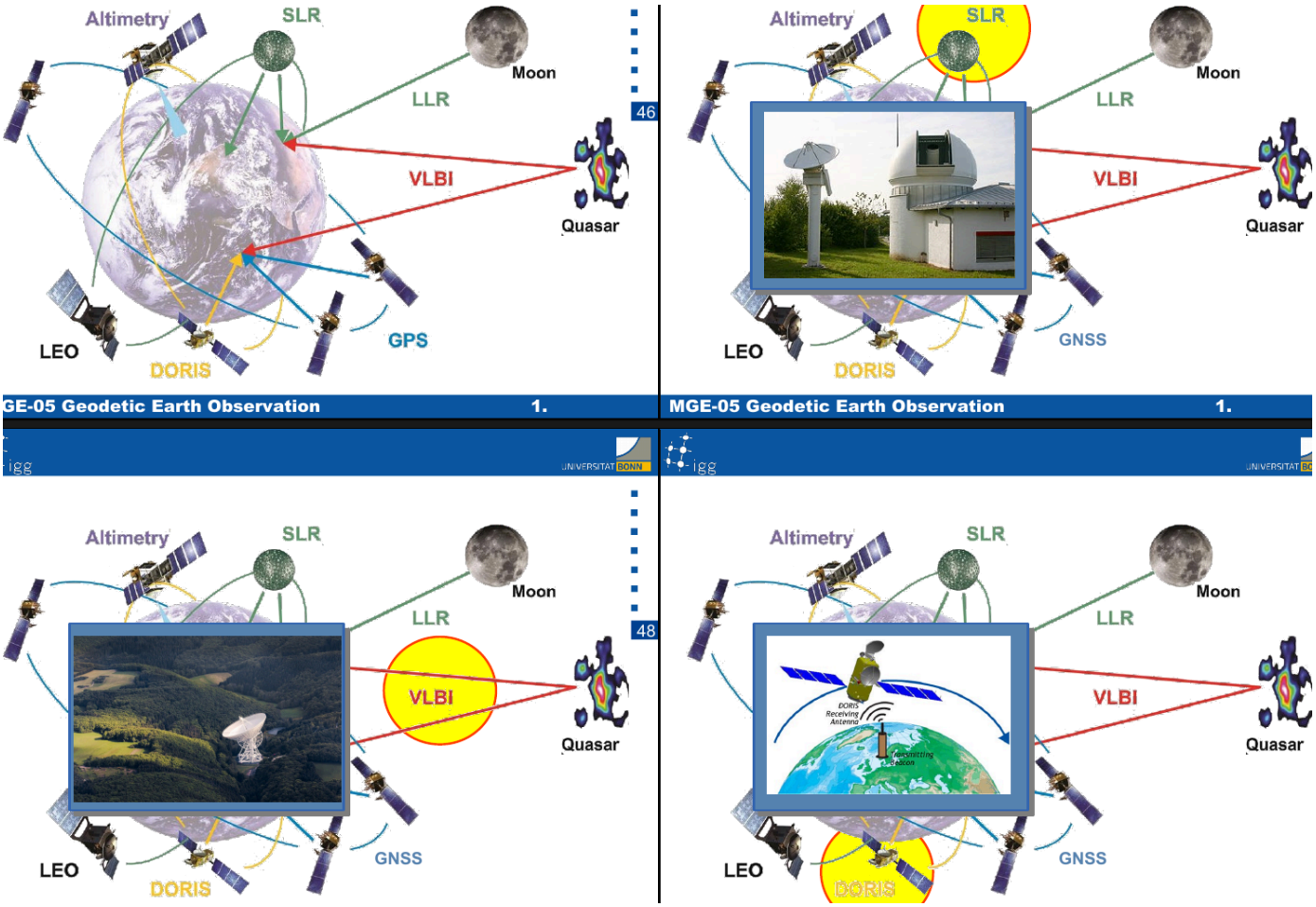
A Realisation: GRS80

$$\left.\begin{aligned} a &= 6378137m \\ J_2 &= 108263 \times 10^{-8} \\ GM &= 3986005 \times 10^{8} m^3 s^{-2} \\ \Omega &= 7292115 \times 10^{-11} rad/s \end{aligned}\right\}$$

All other parameters can be derived from the 4 defining parameters!

Space Geodetic Techniques

- Levelling
- Terrestrial Gravimetry
- Tide gauges
- Pendulum
- Tilt measurements



Summary

- 3 Pillar Definition of geodesy
  - Shape
  - Gravity
  - Orientation
- Changes are linked to geophysical phenomena
  - e.g. ice ages
  - Changes in length of day
  - Surface deformation
- Geodetic observations are needed to realise reference frames and used to understand the Earth system
- Needed: Concepts to link observations to the 3 pilars of geodesy